1 AC input, protection, relays, mains sensing

File: s1_1_ac_input__protection__relays.kicad_sch

2 Rectifier and PFC (CCM, NCP1653, 390 V)

File: s2_2_rectifier_and_pfc__ccm__ncp1.kicad_sch

3 Auxiliary supply (VIPer26 flyback, 1500 V pre-regulator)

File: s3_3_auxiliary_supply__viper26_fl.kicad_sch

4 Inverter (discrete: 3 x IRS2186 + 6 x TO-247 MOSFET), 3-shunt sensing

File: s4_4_inverter__discrete__3_x_irs2.kicad_sch

5 MCU STM32G431CBT6 (hot side)

File: s5_5_mcu_stm32g431cbt6__hot_side.kicad_sch

6 Relay drivers, pump and drain outputs (220 VAC)

File: s6_6_relay_drivers__pump_and_drai.kicad_sch

7 Isolated side: isolator, PHY MCU, 2-wire panel bus, RS-485 to battery module

File: s7_7_isolated_side__isolator__phy.kicad_sch

8 OPTION battery module (DNP in the mains-only version), mechanics, test points

File: s8_8_option_battery_module__dnp_i.kicad_sch

DANGER: mains-referenced except the SELV area (panel bus, RS-485)

Generated from hardware/v2/schematic.py — validated against KiCad 7.0.11 libs and ST pin data

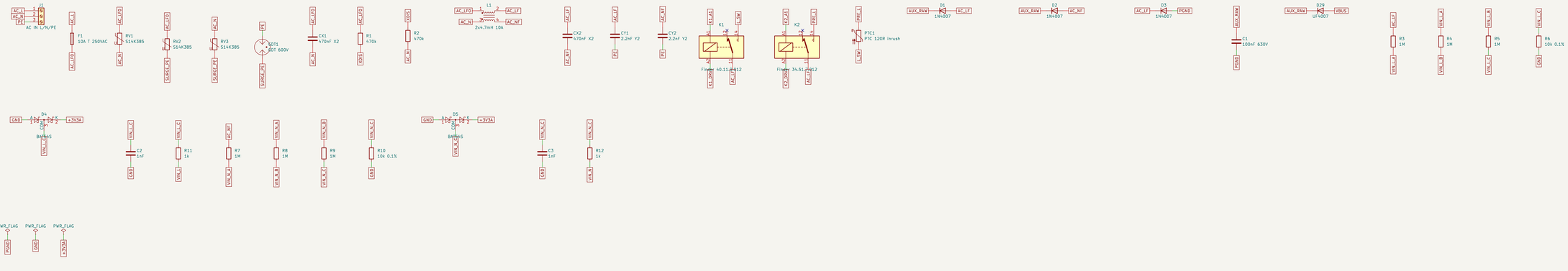
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Sheet: /

File: bldc_cooler_v2.kicad_sch

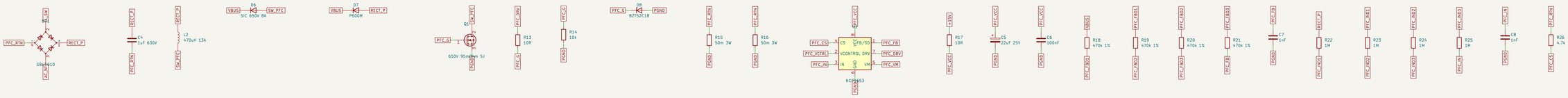
Title: BLDC cooler motor drive v2 (220 VAC, PFC, sensorless FOC)

Size: A3	Date: 2026-11-06	**Rev: 2.0**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/9



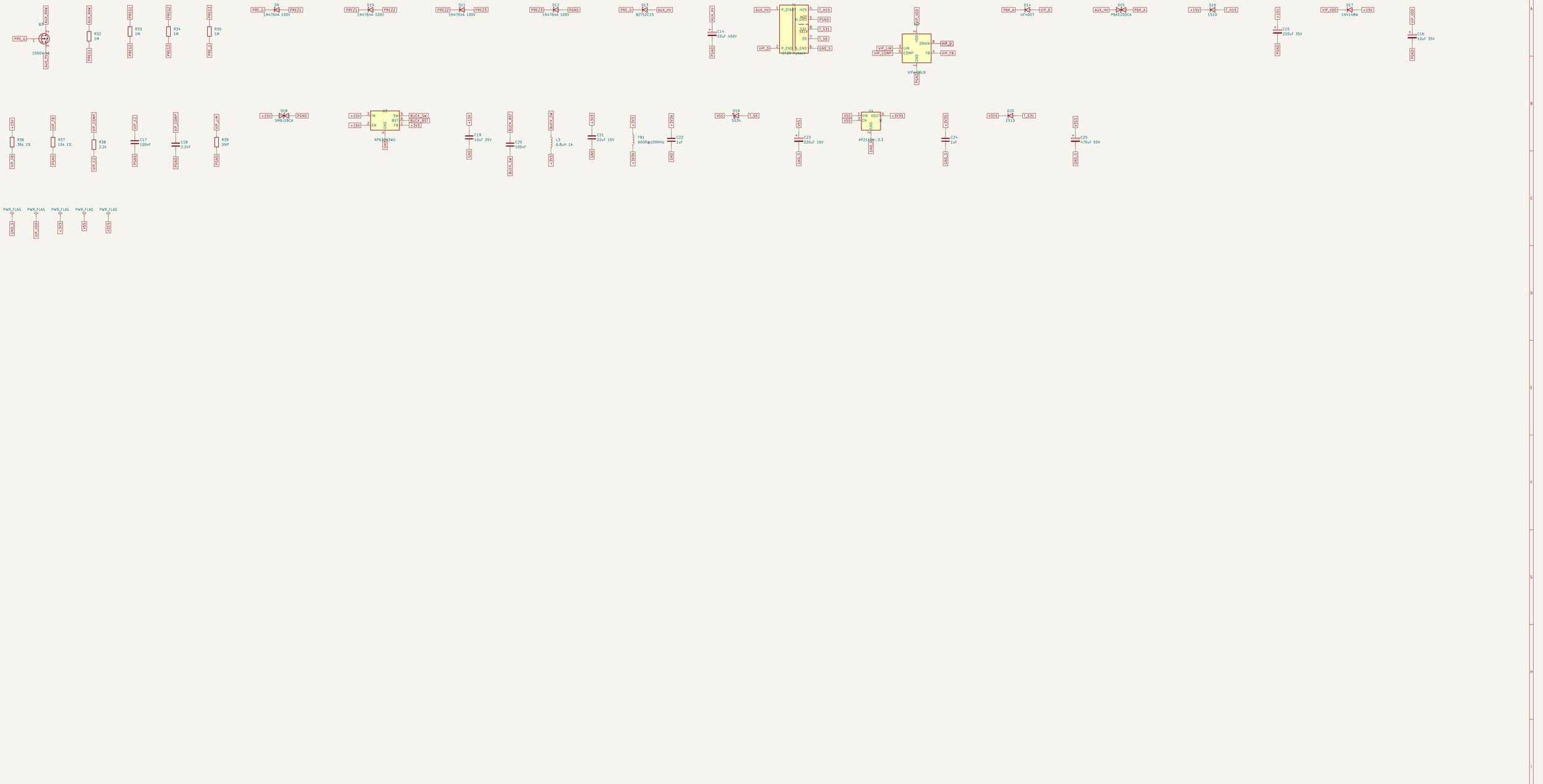
DANGER: mains-referenced except the SELV area (panel bus, RS-485)
Generated from hardware/v2/schematic.py — validated against KiCad 7.0.11 libs and ST pin data
Multiact

Sheet: /1 AC input, protection, relays, mains sensing/	
File: s1_1_ac_input__protection__relays.kicad_sch	
Title: 1 AC input, protection, relays, mains sensing	
Size: A1 — Date: 2026-11-06	**Rev: 2.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 2/9



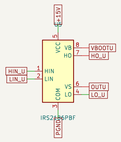
DANGER: mains-referenced except the SELV area (panel bus, RS-485)
Generated from hardware/v2/schematic.py — validated against KiCad 7.0.11 libs and ST pin data
Multiact

Sheet: /2 Rectifier and PFC (CCM, NCP1653, 390 V)/		
File: s2_2_rectifier_and_pfc_ccm_ncp1.kicad_sch		
Title: 2 Rectifier and PFC (CCM, NCP1653, 390 V)		
Size: A1	Date: 2026-11-06	**Rev: 2.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/9



DANGER: mains-referenced except the SELV area (panel bus, RS-485)
Generated from hardware/v2/schematic.py — validated against KiCad 7.0.11 libs and ST pin data
Multiact

Sheet: /3 Auxiliary supply (ViPer26 flyback, 1500 V pre-regulator)/	
File: s3_3_auxiliary_supply__viper26_fl.kicad_sch	
Title: 3 Auxiliary supply (ViPer26 flyback, 1500 V pre-regulator)	
Size: A1 — Date: 2026-11-06	**Rev: 2.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 4/9

















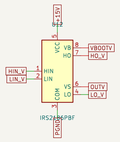


























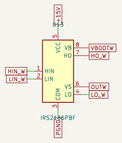






















































































































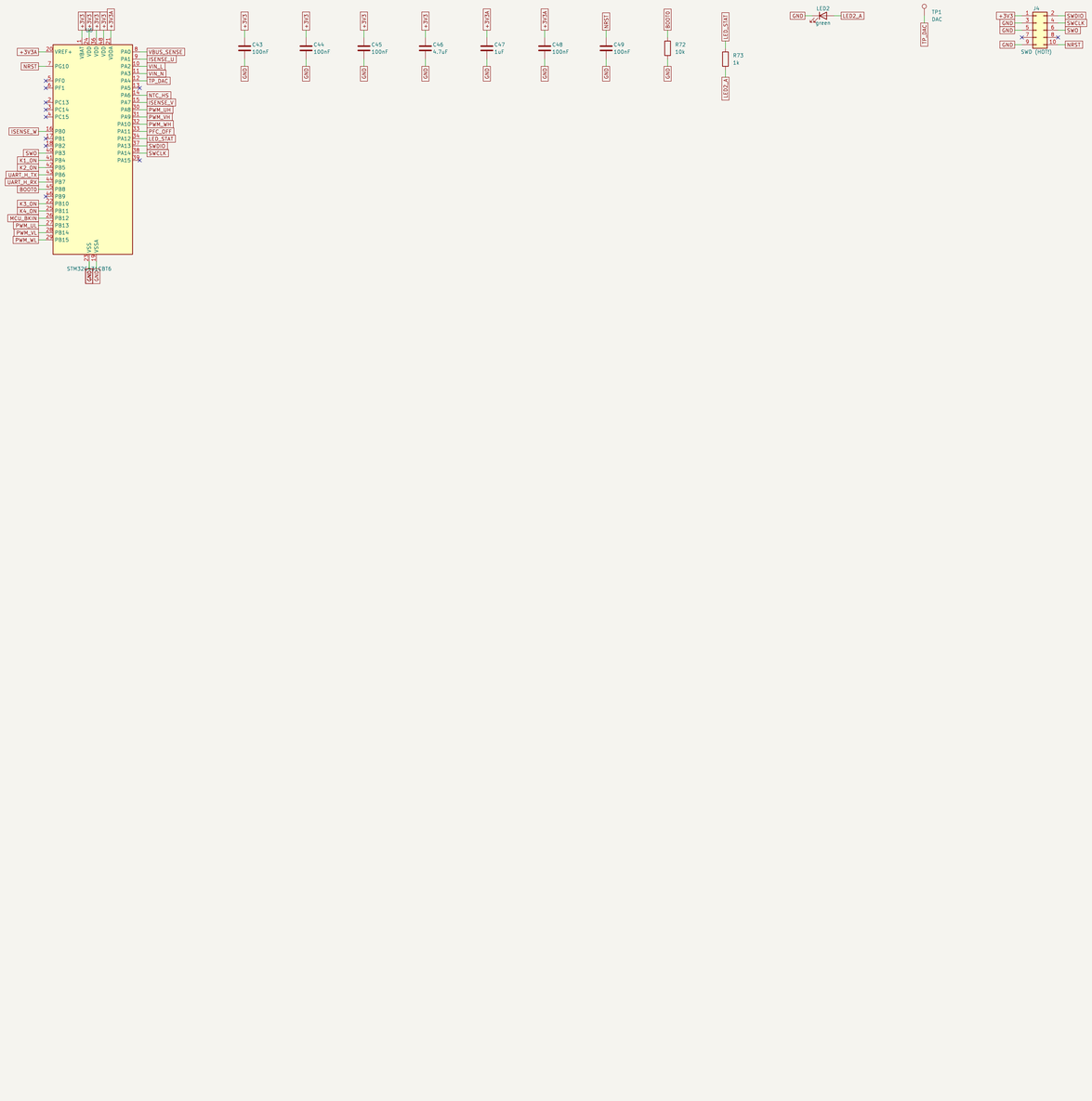






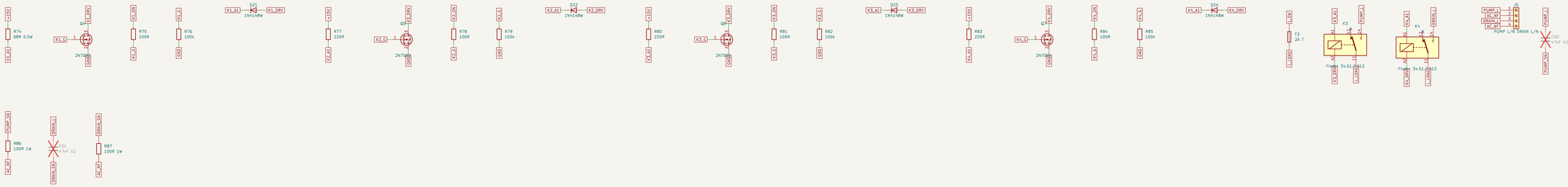
DANGER: mains-referenced except the SELV area (panel bus, RS-485)
Generated from hardware/v2/schematic.py — validated against KiCad 7.0.11 libs and ST pin data
Multiact

Sheet: /4 Inverter (discrete: 3 x IRS2186 + 6 x TO-247 MOSFET), 3-shunt sensing/	
File: s4_4_inverter_discrete_3_x_irs2.kicad_sch	
Title: 4 Inverter (discrete: 3 x IRS2186 + 6 x TO-247 MOSFET), 3-shunt sensing	
Size: A1	Date: 2026-11-06
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 5/9



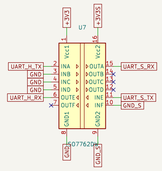
DANGER: mains-referenced except the SELV area (panel bus, RS-485)
Generated from hardware/v2/schematic.py — validated against KiCad 7.0.11 libs and ST pin data
Multiact

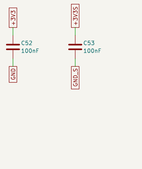
Sheet: /5 MCU STM32G431CBT6 (hot side)/		
File: s5_5_mcu_stm32g431cbt6_hot_side.kicad_sch		
Title: 5 MCU STM32G431CBT6 (hot side)		
Size: A1	Date: 2026-11-06	**Rev: 2.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/9

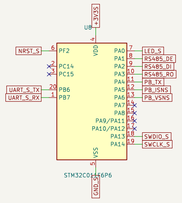


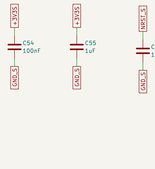
DANGER: mains-referenced except the SELV area (panel bus, RS-485)
Generated from hardware/v2/schematic.py — validated against KiCad 7.0.11 libs and ST pin data
Multiact

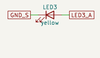
Sheet: /6 Relay drivers, pump and drain outputs (220 VAC)/	
File: s6_6_relay_drivers__pump_and_drai.kicad_sch	
Title: 6 Relay drivers, pump and drain outputs (220 VAC)	
Size: A1 — Date: 2026-11-06	**Rev: 2.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 7/9



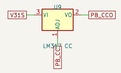


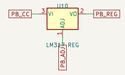














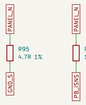


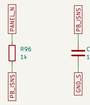


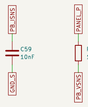


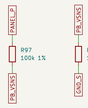






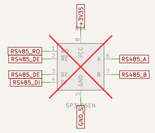




















DANGER: mains-referenced except the SELV area (panel bus, RS-485)
Generated from hardware/v2/schematic.py — validated against KiCad 7.0.11 libs and ST pin data
Multiact

Sheet: /7 Isolated side: isolator, PHY MCU, 2-wire panel bus, RS-485 to battery module/	
File: s7_7_isolated_side__isolator_phy.kicad_sch	
Title: 7 Isolated side: isolator, PHY MCU, 2-wire panel bus, RS-485 to batte	
Size: A1	Date: 2026-11-06 **Rev: 2.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 8/9



DANGER: mains-referenced except the SELV area (panel bus, RS-485)
Generated from hardware/v2/schematic.py — validated against KiCad 7.0.11 libs and ST pin data
Multiact

Sheet: /8 OPTION battery module (DNP in the mains-only version), mechanics, test points/	
File: s8_8_option_battery_module__dnp_i.kicad_sch	
Title: 8 OPTION battery module (DNP in the mains-only version), mechanics,	
Size: A1	Date: 2026-11-06 **Rev: 2.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 9/9